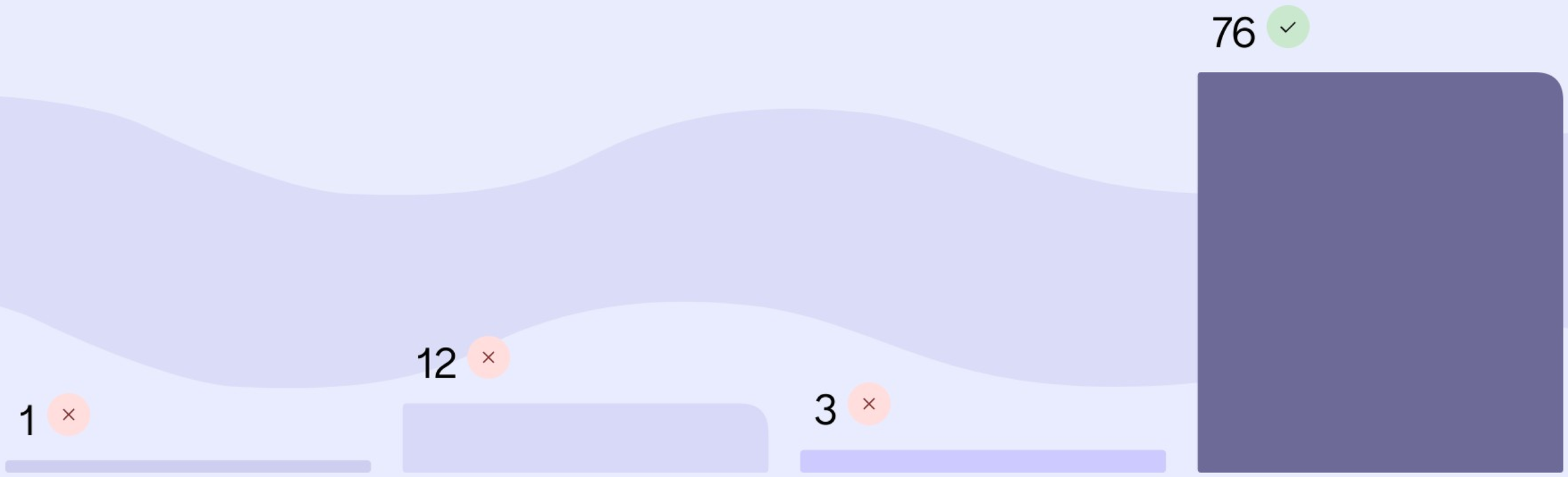
# Example questions HCI exam

Chapter 1 example

# In which period did the computer mouse become available outside the laboratory?




In the 1940s as part of the memex invitend by Vannevar Bush

In the early 1960s as part of the Engelbert's NLS

In the late 1960s as part of Sunderland's Sketchpad

In the early 1980s as part of the Xerox Star system

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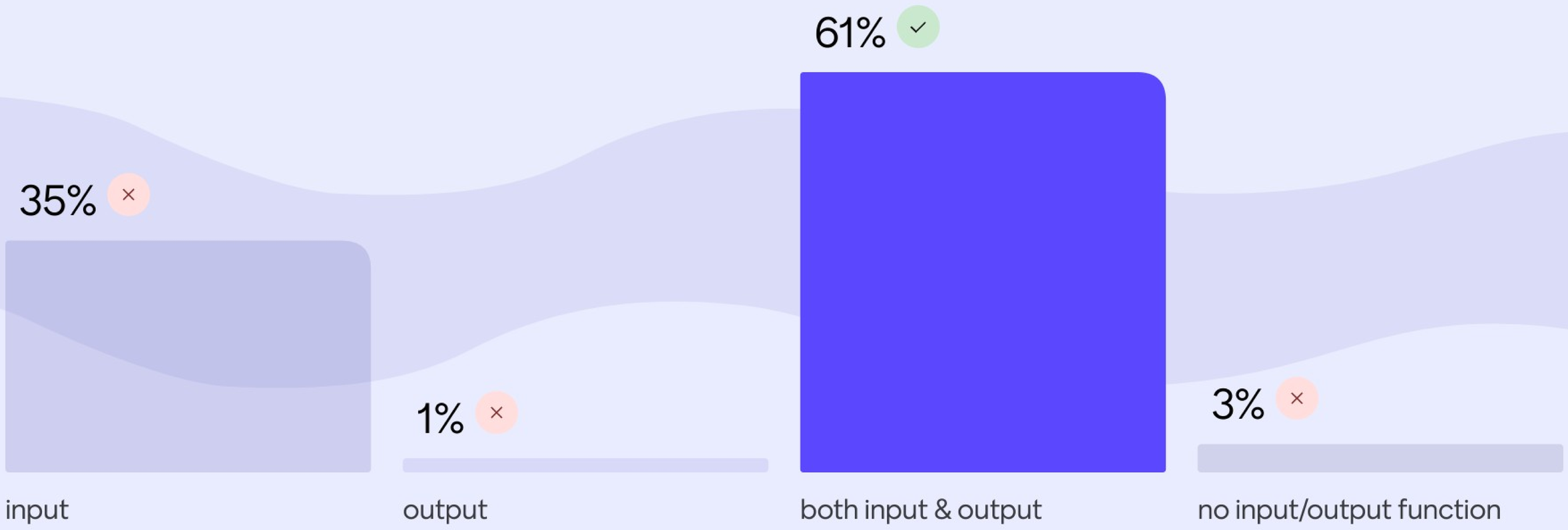
Chapter 2 example

# In an eye-tracking typing system for people unable to use their hands, what is the function of the eye?

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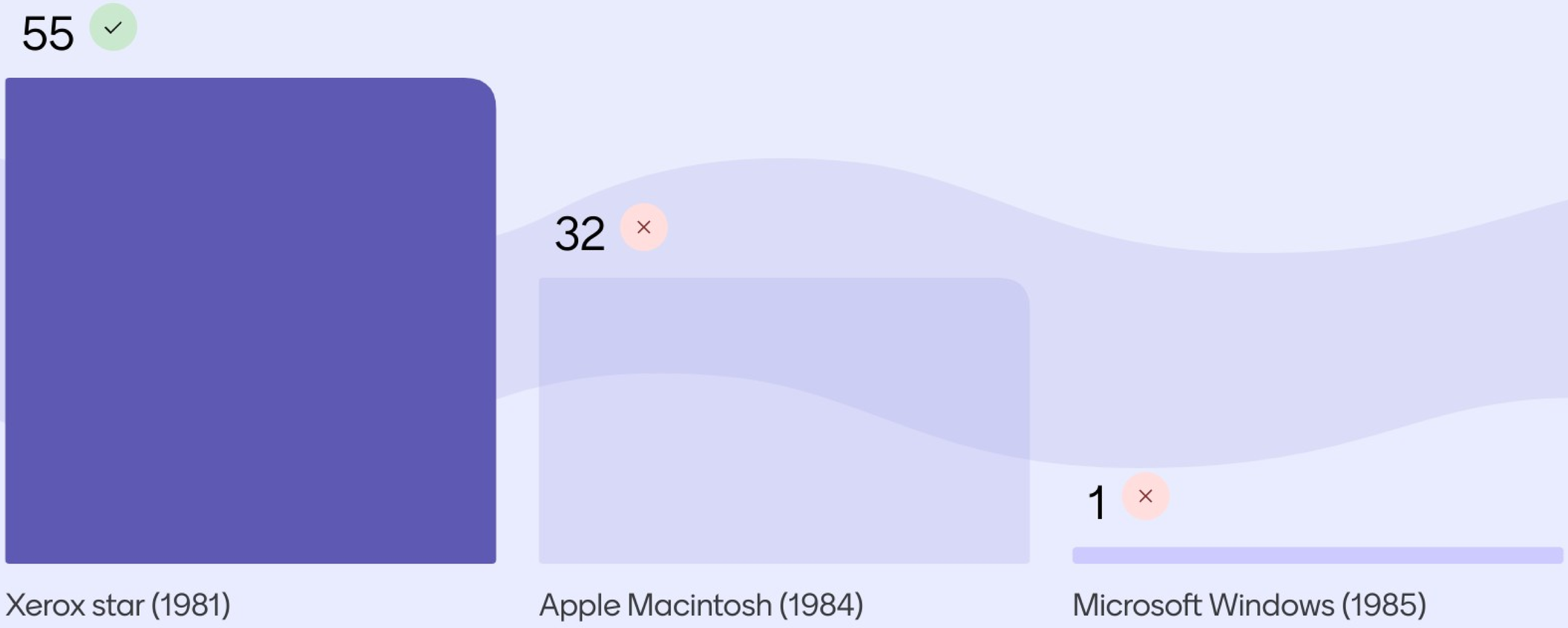
# The first commercially available computer system to use a GUI was

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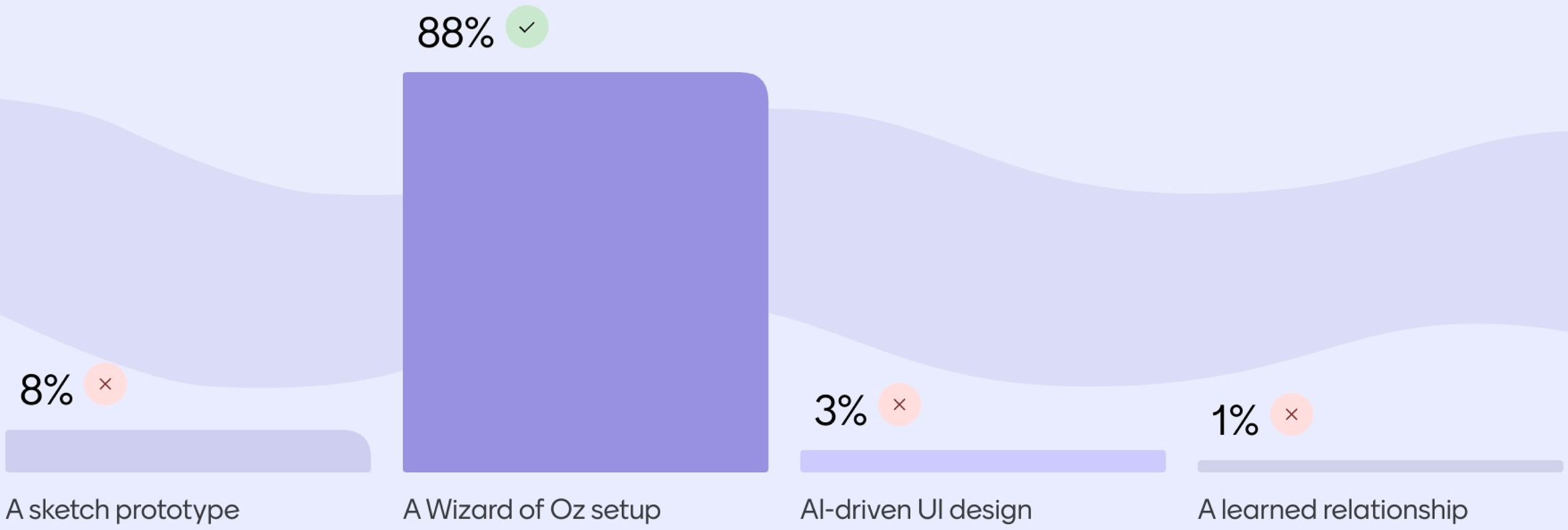
Week 4 lectures & slides example:

A company is testing the UI of an AI chatbot. They have not built the AI yet, they test with answers provided by humans instead. This setup is

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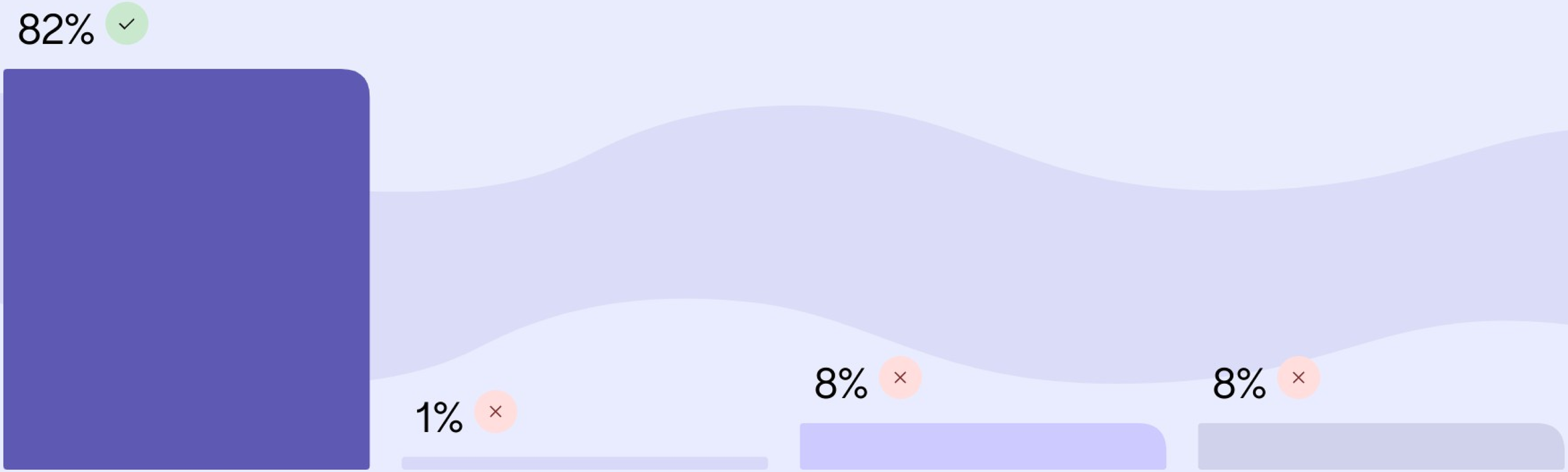
Chapter 5 example question

We study the effect of adding colored buttons on usability of a device. We test with a colored and black version and measure effect with the SUS.




In the experiment, button color is the independent variable

In the experiment, usability is an independent variable

In the experiment, button color is the dependent variable

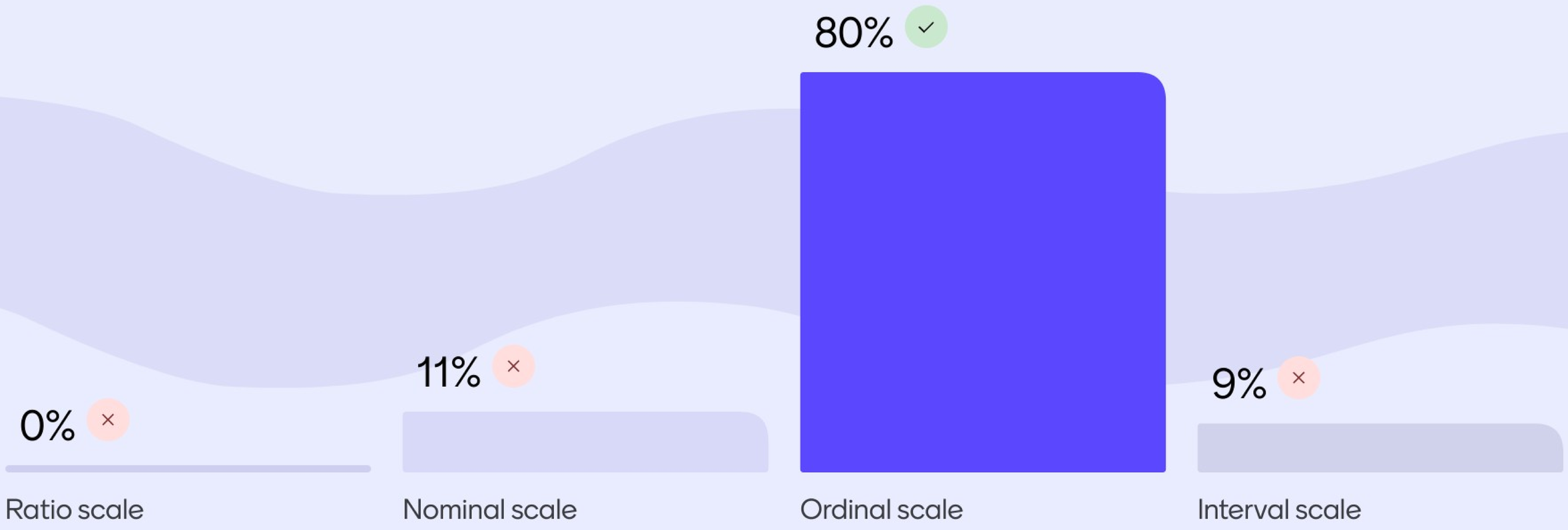
In the experiment, the task completion time is the dependent variable

# Consider the following question on a questionnaire; How would you rate the current temperature in your room?

- way too cold
- slightly cold
- good
- slightly warm
- too hot
- inferno!

Chapter 4 example

# The answer to the question on the previous slide is an example of

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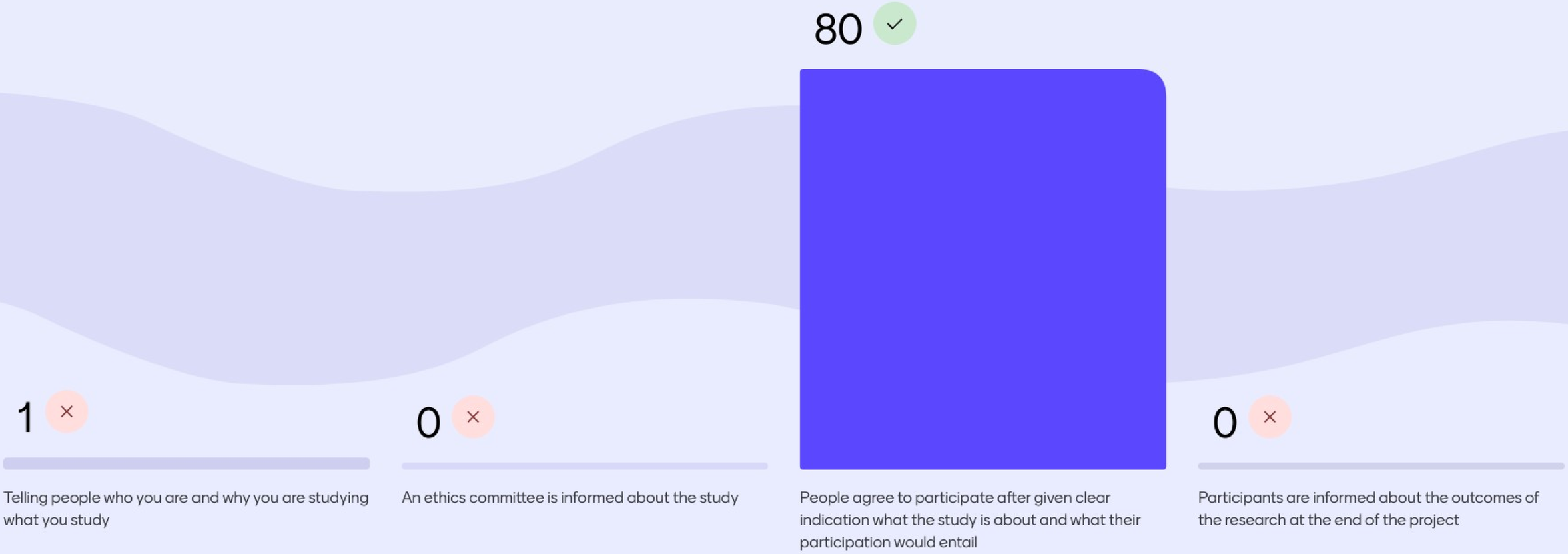
Chapter 5 example question

# What is meant by ‘informed consent’ in human subject experiments? Please indicate the best answer.

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Chapter 6 example

A group of Android users (A) and iPhone users (B) are asked how satisfied they are with their phone. The experiment uses a 5 point Likert-scale. What is the most appropriate test to see if the answers between group A and B differ significantly?

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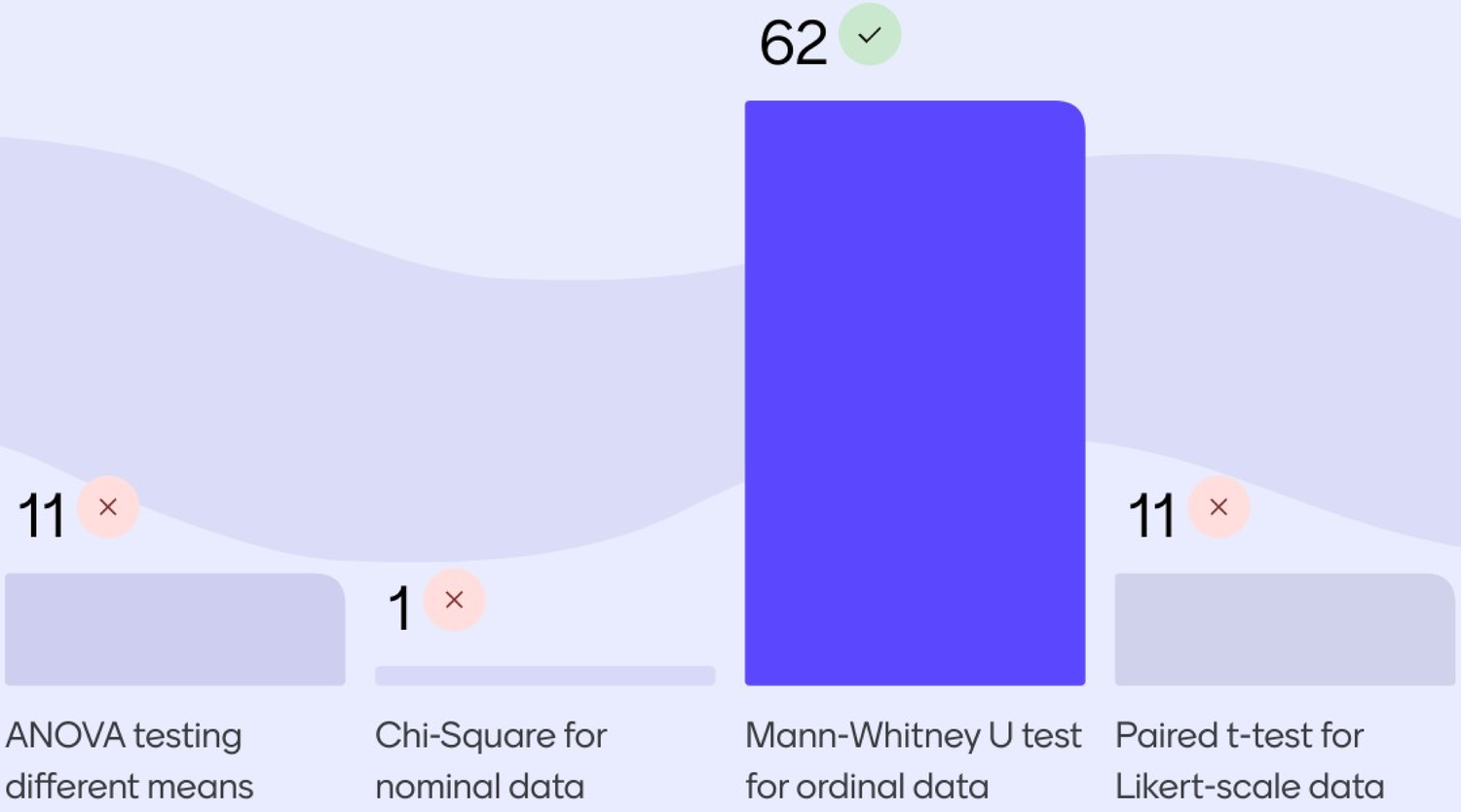
| Data Setup | Parametric test | Non-Parametric test |
|---|---|---|
| 1 Variable<br>2 Categories<br>Between Subjects | independent t-test | Mann-Whitney U test |
| 1 Variable<br>2 Categories<br>Within-Subjects | paired t-test | Wilcoxon Signed Rank Test |
| 1 Variable<br>>2 Categories<br>Between Subjects | One-way ANOVA | Kruskal Wallis Test |
| 1 Variable<br>>2 Categories<br>Within Subjects | repeated measures ANOVA | Friedman test<br>Mood's median test |
| 1 Variable<br>(Correlation) | Pearson's r | Spearman's ρ (rho) |

Chapter 6 example




| rametric test | N |
|---|---|
| dependent t-test | |
| paired t-test | W |
| ne-way ANOVA | |
| peated measures ANOVA | |
| Pearson's r | |

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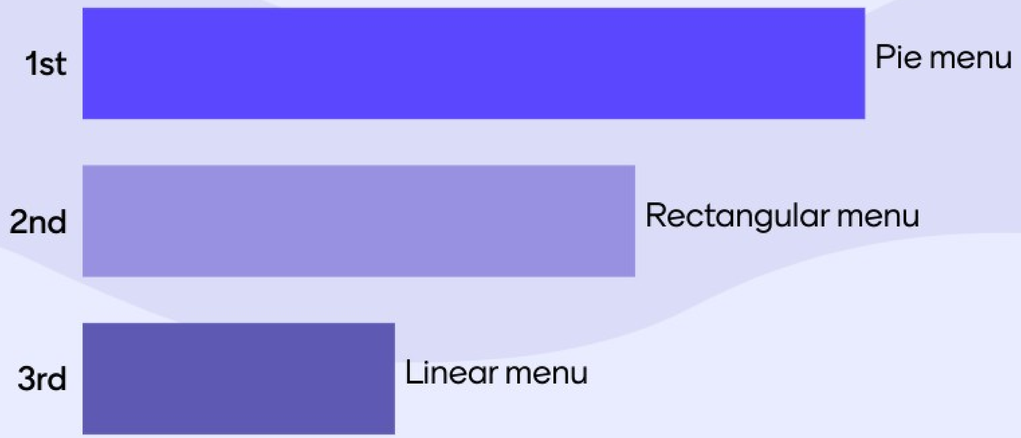
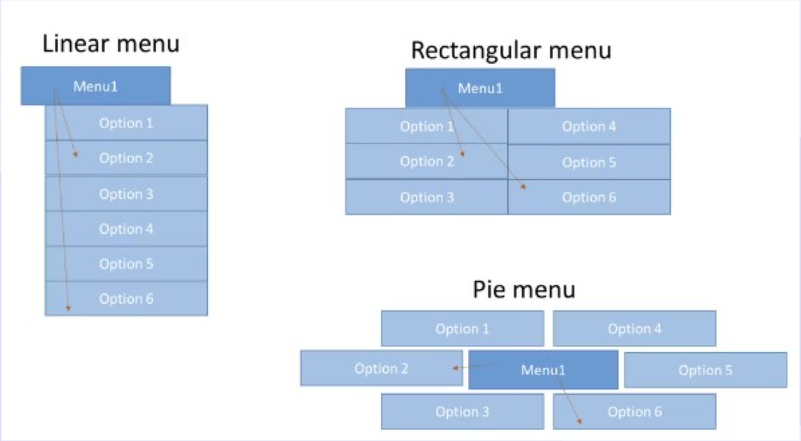
chapter 7 example

Order the three menus, where the options are shown after the user clicks “Menu1", from fastest to slowest , according to Fitts' law:



chapter 7 example

# Compute KLM-predicted task time for an icon deletion task sequence:

1. initiate the deletion
2. find the file icon
3. point to file icon
4. press and hold mouse
5. drag file to trash can
6. release mouse
7. point to original window

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| Operator | Description | Associated Time |
|---|---|---|
| ***K*** | Keystroke, typing one letter, number, etc. or function key like 'CTRL', 'SHIFT' | Expert typist (90 wpm): 0.12 sec<br>Average skilled typist (55 wpm): 0.20 sec<br>Average non-secretarial typist (40 wpm): 0 .28 sec<br>Worst typist (unfamiliar with keyboard): 1.2 sec |
| ***H*** | 'Homing', moving the hand between mouse and keyboard | 0.4 sec |
| ***B / BB*** | Pressing / clicking a mouse button | 0.1 sec / 2*0.1 sec |
| ***P*** | Pointing with the mouse to a target | 0.8 to 1.5 sec with an average of 1.1 sec<br>Can also use Fitts' Law |
| $D(n_D, l_D)$ | Drawing $n_D$ straight line segments of length $l_D$ | $0.9*n_D + 0.16*l_D$ |
| ***M*** | Subsumed time for mental acts; sometimes used as 'look-at' | 1.35 sec (1.2 sec according to [Olson and Olson 1995]) |
| ***R(t)*** or ***W(t)*** | System response (or 'work') time, time during which the user cannot act | Dependent on the system, to be determined on a system-by-system basis |

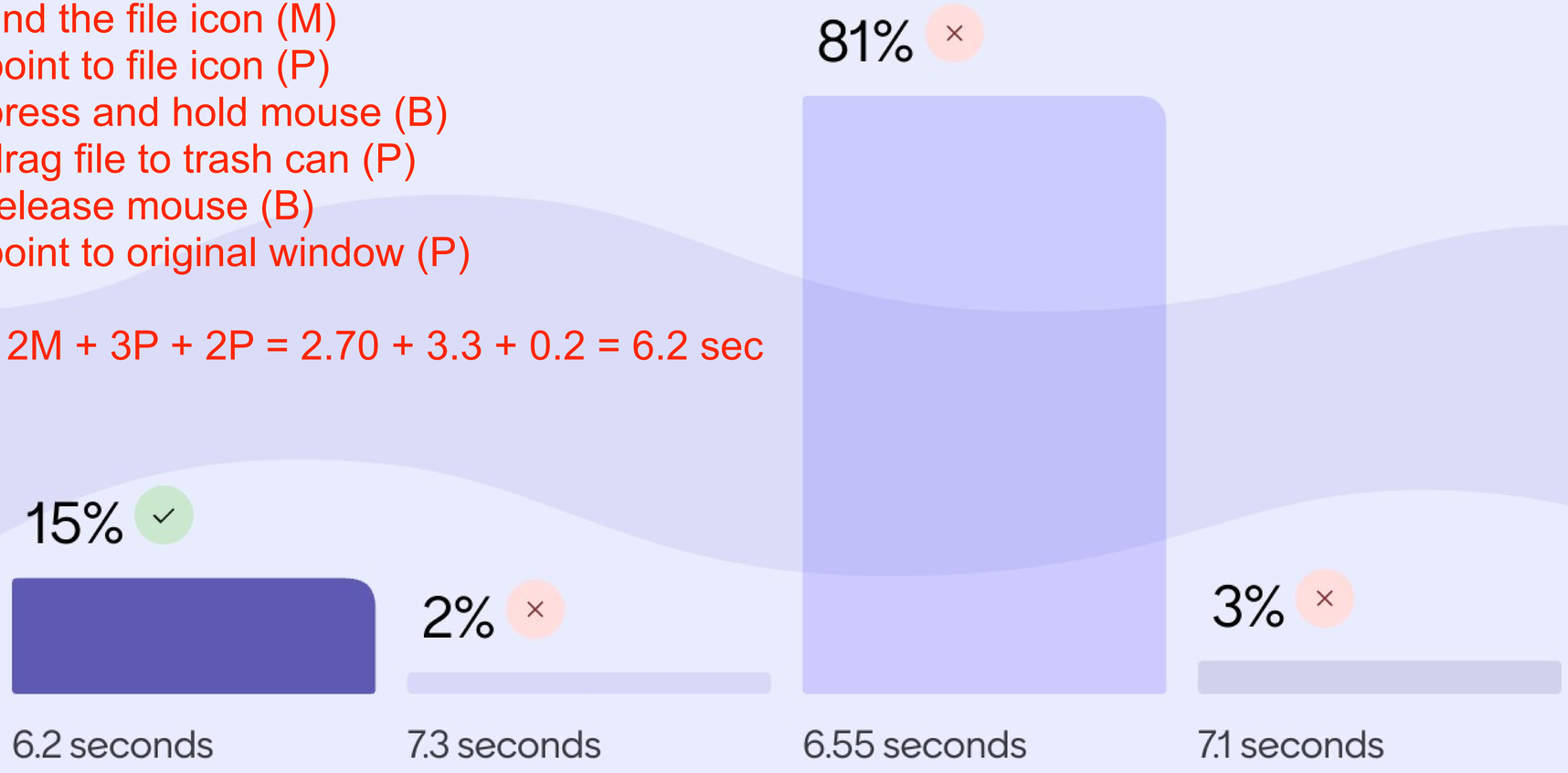
| | Associated |
|---|---|
| ter, number, CTRL’, | Expert typi<br>Average sk<br>Average no<br>Worst typis |
| nd between | 0.4 sec |
| se button | 0.1 sec / 2* |
| to a target | 0.8 to 1.5 s<br>Can also us |
| segments of | $0.9*n_D + 0.$ |
| al acts; at’ | 1.35 sec (1. 1995]) |
| k’) time, er cannot act | Dependent system-by- |

Mentimeter

# Multiple Choice

Solution:
1. initiate the deletion (M)
2. find the file icon (M)
3. point to file icon (P)
4. press and hold mouse (B)
5. drag file to trash can (P)
6. release mouse (B)
7. point to original window (P)

T = 2M + 3P + 2P = 2.70 + 3.3 + 0.2 = 6.2 sec

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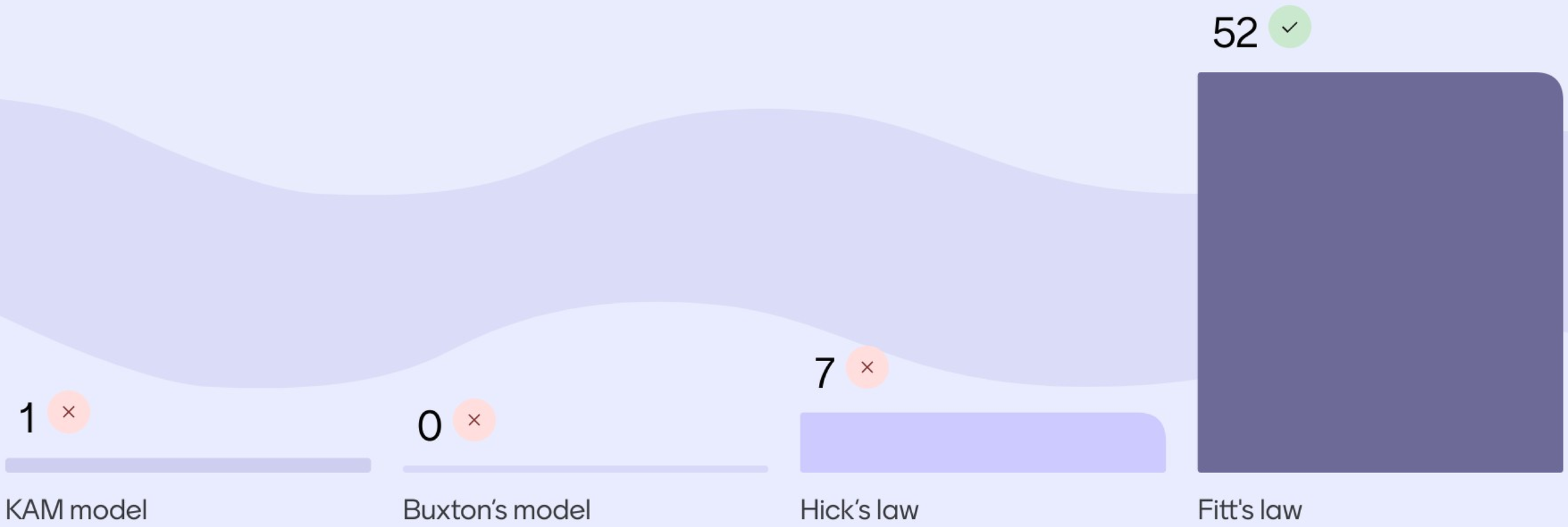
Chapter 7

What HCI model or law states that the time it takes a user to acquire a target is a function of the size of the target and the distance to the target?

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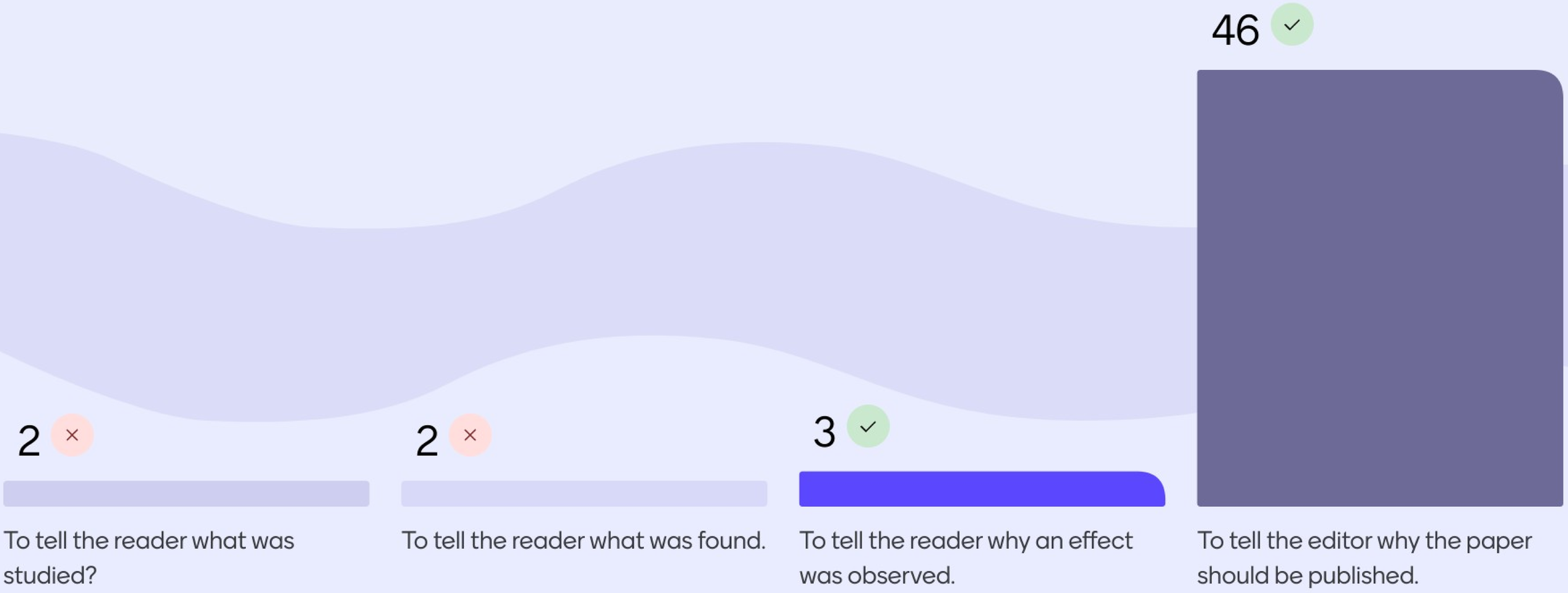
# Which of the following are NOT the goals of an abstract section in a research paper?

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